

Topic 9: Computing the Spectrum

[Summary based on J. H. McClellan, R. W. Schafer, and M. A. Yoder, *Signal Processing First*, Prentice Hall, 2003.]

Topics: DFT, FFT, windowing, spectral leakage, resolution, zero-padding, periodic signals, and spectrograms.

Key idea. A computer does not evaluate the continuous-time Fourier-transform integral directly. It estimates a spectrum from sampled, finite-duration data, normally by computing a DFT efficiently with an FFT algorithm.

9.1 From the Fourier integral to a finite Fourier sum

For a continuous-time signal $x_c(t)$, the continuous-time Fourier transform (CTFT) is

$$X_c(j\omega) = \int_{-\infty}^{\infty} x_c(t) e^{-j\omega t} dt.$$

After sampling with period T_s , so that $x[n] = x_c(nT_s)$, the integral can be approximated by a sum. For a right-sided signal and a finite set of L samples,

$$\hat{X}_c(j\omega_k) = T_s \sum_{n=0}^{L-1} x[n] e^{-j\omega_k n T_s}.$$

Choosing N equally spaced frequencies over one period gives

$$\omega_k = \frac{2\pi k}{NT_s}, \quad k = 0, 1, \dots, N-1.$$

Substitution gives the finite Fourier sum

$$X[k] = \sum_{n=0}^{L-1} x[n] e^{-j(2\pi/N)kn}, \quad k = 0, 1, \dots, N-1.$$

When $L = N$, the DFT maps N time samples to N complex frequency coefficients and has an exact inverse.

9.2 How the Fourier transforms are related

	Discrete time	Continuous time
Discrete frequency	DFT, $X[k]$	Fourier series, $\{a_k\}$
Continuous frequency	DTFT, $X(e^{j\hat{\omega}})$	CTFT, $X_c(j\omega)$

All four representations use the same underlying operation: correlating or projecting a signal onto complex exponentials.

9.2.1 DTFT of a sampled signal

The discrete-time Fourier transform is

$$X(e^{j\hat{\omega}}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\hat{\omega}n}.$$

It is continuous in normalized frequency $\hat{\omega}$ and periodic with period 2π . If $x[n] = x_c(nT_s)$, sampling creates shifted copies of the CTFT:

$$X(e^{j\omega T_s}) = \frac{1}{T_s} \sum_{\ell=-\infty}^{\infty} X_c(j(\omega - \ell\omega_s)), \quad \omega_s = \frac{2\pi}{T_s}.$$

For a bandlimited signal sampled above the Nyquist rate, the copies do not overlap. In the baseband,

$$X(e^{j\omega T_s}) = \frac{1}{T_s} X_c(j\omega), \quad |\omega| < \frac{\pi}{T_s}.$$

9.2.2 DFT as samples of the DTFT

For a finite sequence, the DFT is obtained by evaluating its DTFT at N equally spaced normalized frequencies:

$$X[k] = X(e^{j2\pi k/N}), \quad k = 0, 1, \dots, N-1.$$

Three effects must be kept separate: sampling makes the spectrum periodic, windowing changes its shape, and the DFT samples that changed spectrum on a discrete grid.

9.3 Time windowing and spectral leakage

A DFT requires a finite sequence. Selecting a finite segment is equivalent to multiplying the full signal by a data window:

$$\tilde{x}[n] = w[n]x[n].$$

A length- L rectangular window is

$$w_R[n] = \begin{cases} 1, & 0 \leq n \leq L-1, \\ 0, & \text{otherwise.} \end{cases}$$

In continuous time, a duration- T rectangular window has the transform

$$W_R(j\omega) = \frac{\sin(\omega T/2)}{\omega/2} e^{-j\omega T/2}.$$

Its central lobe lies between the first zeros and has approximate width

$$\Delta\omega_{\text{main}} \approx \frac{4\pi}{T}.$$

Multiplication in time corresponds to convolution in frequency. Each ideal spectral line is therefore replaced by a shifted, scaled copy of the window spectrum. This spreading is called **spectral leakage**. Longer windows produce narrower mainlobes and better frequency resolution.

9.3.1 Rectangular versus Hamming windows

The rectangular window has a relatively narrow mainlobe but high sidelobes. The continuous-time Hamming window is

$$w_H(t) = \begin{cases} 0.54 - 0.46\cos\left(\frac{2\pi t}{T}\right), & 0 \leq t \leq T, \\ 0, & \text{otherwise.} \end{cases}$$

The Hamming window has much lower sidelobes, reducing misleading low-level peaks, but its mainlobe is about twice as wide. For comparable frequency resolution, it therefore requires approximately twice the duration of a rectangular window.

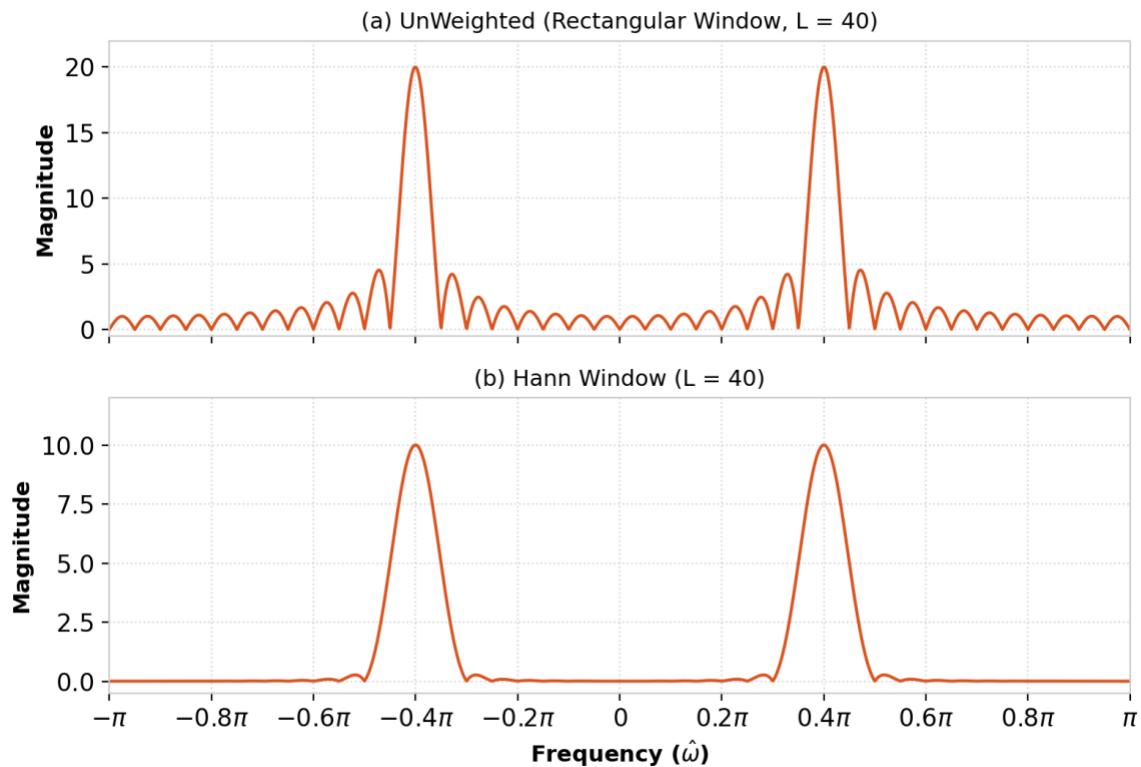


Figure 9.1. Effect of window shape on the spectrum of a windowed sinusoid.

This slide uses a Hann window (very similar in purpose to a Hamming window) and compares it with a rectangular window of the same length. The rectangular window has a narrower mainlobe but stronger sidelobes, whereas the Hann/Hamming-type window greatly suppresses sidelobes at the cost of a wider mainlobe.

9.3.2 Resolution trade-off

A long window improves frequency resolution but blurs rapid changes in time. A short window improves time localization but merges nearby frequencies. Time and frequency resolution cannot be improved simultaneously.

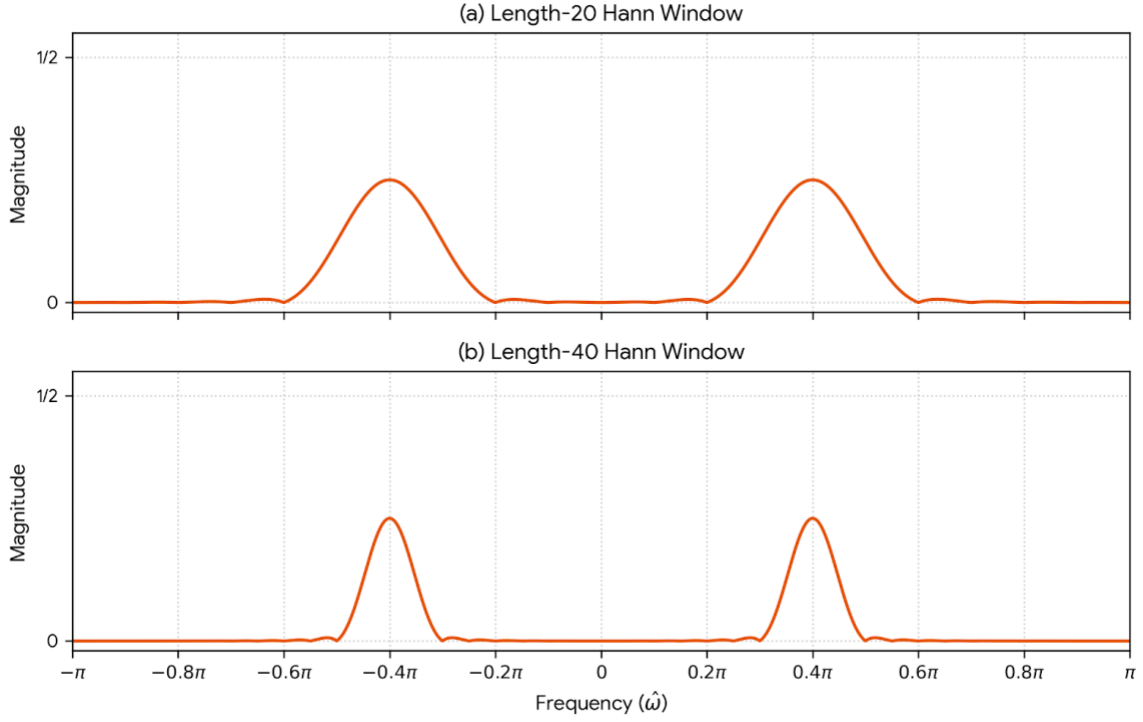


Figure 9.2. Effect of window length on frequency resolution.

With the same window type, a longer window produces a narrower mainlobe and therefore better frequency resolution. The shorter window gives poorer frequency resolution but better time localization.

9.4 Spectrum of a windowed sum of sinusoids

Consider a DC term plus one sinusoid:

$$x_c(t) = A_0 + A_1 \cos(\omega_1 t + \phi_1).$$

With $X_1 = A_1 e^{j\phi_1}$, its ideal CTFT is

$$X_c(j\omega) = 2\pi A_0 \delta(\omega) + \pi X_1 \delta(\omega - \omega_1) + \pi X_1^* \delta(\omega + \omega_1).$$

After multiplication by a window $w_c(t)$,

$$\tilde{X}_c(j\omega) = A_0 W_c(j\omega) + \frac{1}{2} X_1 W_c(j(\omega - \omega_1)) + \frac{1}{2} X_1^* W_c(j(\omega + \omega_1)).$$

The same principle applies to every sinusoid in a sum. Distinct peaks are visible only when the shifted mainlobes are sufficiently separated.

9.5 The DFT, inverse DFT, and FFT

For an N -sample sequence, the forward and inverse DFT form an exact analysis-synthesis pair:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}, \quad k = 0, 1, \dots, N-1.$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}, \quad n = 0, 1, \dots, N-1.$$

A direct DFT requires on the order of N^2 operations. The FFT computes the same DFT in approximately $N \log_2 N$ operations. Radix-2 algorithms are especially efficient when N is a power of two.

9.5.1 Frequency indexing and negative frequencies

The normalized DFT frequencies are

$$\hat{\omega}_k = \frac{2\pi k}{N}.$$

For a real-valued signal, the DFT coefficients satisfy conjugate symmetry:

$$X[N - k] = X^*[k].$$

The corresponding physical frequencies are

$$f_k = \frac{k}{N} f_s, \quad \omega_k = \frac{2\pi k}{NT_s}.$$

9.5.2 Complex exponential and bin-centred sampling

For

$$x[n] = e^{j(\hat{\omega}_0 n + \phi)},$$

the DFT samples a shifted Dirichlet function. If $\hat{\omega}_0$ is not exactly a DFT-bin frequency, energy appears in several coefficients. If

$$\hat{\omega}_0 = \frac{2\pi k_0}{N},$$

then

$$X[k] = N\delta[k - k_0].$$

This “picket-fence” effect is not an FFT error; it results from sampling the DTFT only at the DFT grid

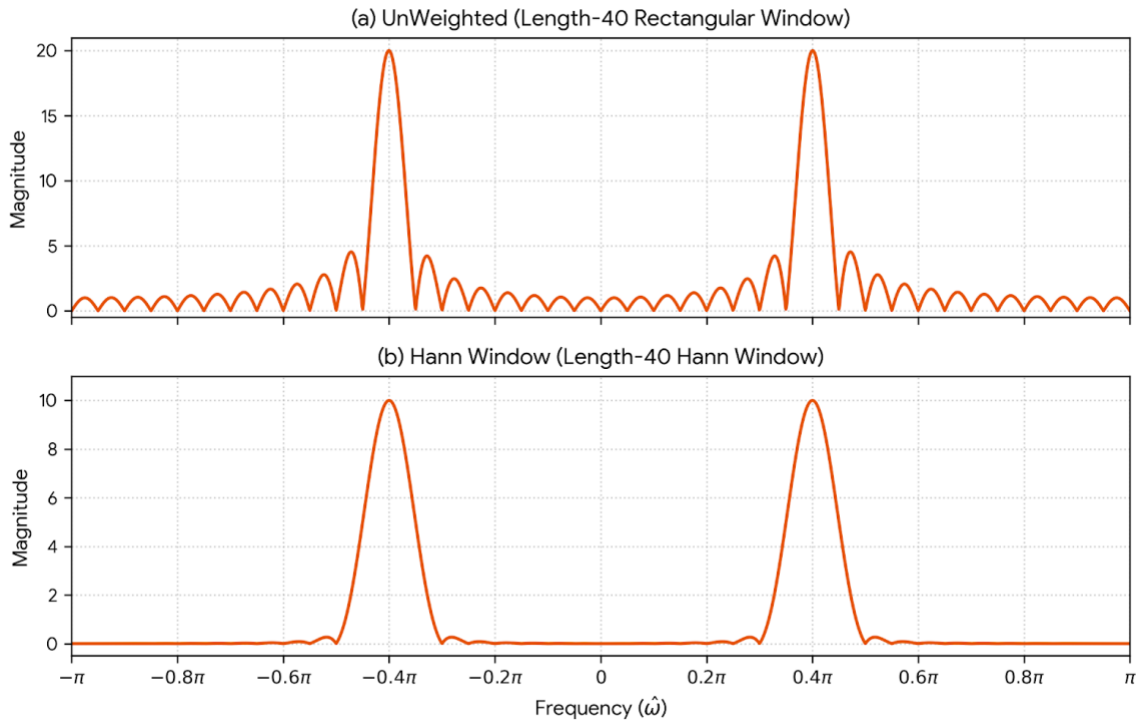


Figure 9.3. DFT of a complex exponential whose frequency is not exactly on a DFT bin.

Because the sinusoid falls between DFT-bin frequencies, the sampled spectrum is spread over several coefficients. The grey envelope is the shifted Dirichlet form, and the red samples show what the DFT actually measures on the frequency grid.

9.6 Finite-length signals and zero-padding

If a signal has L nonzero samples but an N -point DFT is computed with $N > L$, then $N - L$ zeros are appended:

$$X[k] = \sum_{n=0}^{L-1} x[n]e^{-j2\pi kn/N} = X(e^{j2\pi k/N}).$$

Zero-padding evaluates the same DTFT at a denser set of frequencies. It produces a smoother-looking curve and may improve interpolation of peak locations, but it does not narrow the window mainlobe and does not improve true resolving power.

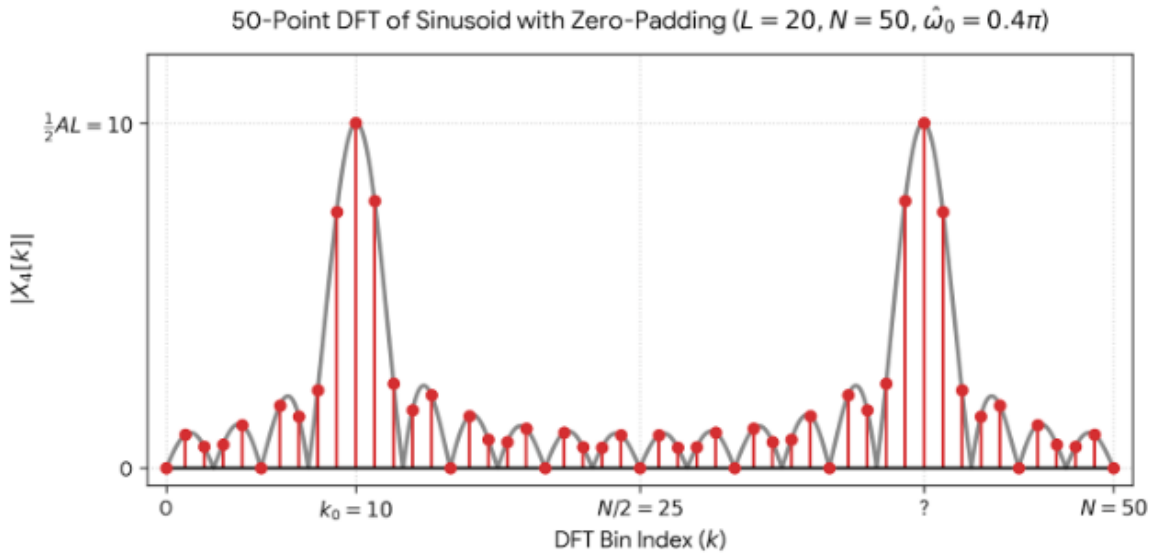


Figure 9.4. Zero-padding gives denser frequency sampling.

Appending zeros before the DFT does not create new spectral information and does not improve the true resolving power. It simply samples the same spectral envelope at more frequency points, making peaks easier to visualize and interpolate.

9.6.1 FIR-filter example

For an FIR filter with impulse response $h[n]$,

$$H(e^{j\hat{\omega}}) = \sum_{n=0}^{L-1} h[n]e^{-j\hat{\omega}n}.$$

An N -point DFT gives sampled values

$$H[k] = \sum_{n=0}^{L-1} h[n]e^{-j2\pi kn/N} = H(e^{j2\pi k/N}).$$

Increasing the physical impulse-response length can sharpen filter features; increasing only N by zero-padding merely draws those features more densely.

9.7 Periodic signals

If the sampled signal has an integer period of N samples and one complete period is transformed, its DFT gives an exact finite set of Fourier-series coefficients. Harmonic components fall exactly on bins when their frequencies are integer multiples of $2\pi/N$. The inverse DFT reconstructs one period, and periodic extension reconstructs the complete sequence.

For periodic analysis, select a record containing an integer number of periods whenever possible. Otherwise, the record boundaries introduce an artificial discontinuity and leakage.

9.8 The spectrogram: spectrum as a function of time

Long signals such as speech and music are rarely stationary over their full duration. A short window is moved along the signal and an FFT is computed at each position:

$$X[k, n] = \sum_{m=0}^{L-1} w[m]x[n+m]e^{-j2\pi km/N}, \quad k = 0, 1, \dots, N-1.$$

The magnitude $|X[k, n]|$, or usually a logarithmic magnitude, is displayed as an image. Time is horizontal, frequency vertical, and intensity or colour represents spectral magnitude. For real signals, only $0 \leq f \leq f_s/2$ normally needs to be shown.

9.8.1 Frequency and time calibration

$$f_k = \frac{k}{N} f_s.$$

Successive frames are separated by a hop size R samples. At sampling rate f_s , the frame time is approximately nR/f_s . Adjacent windows commonly overlap by 50% to 80%.

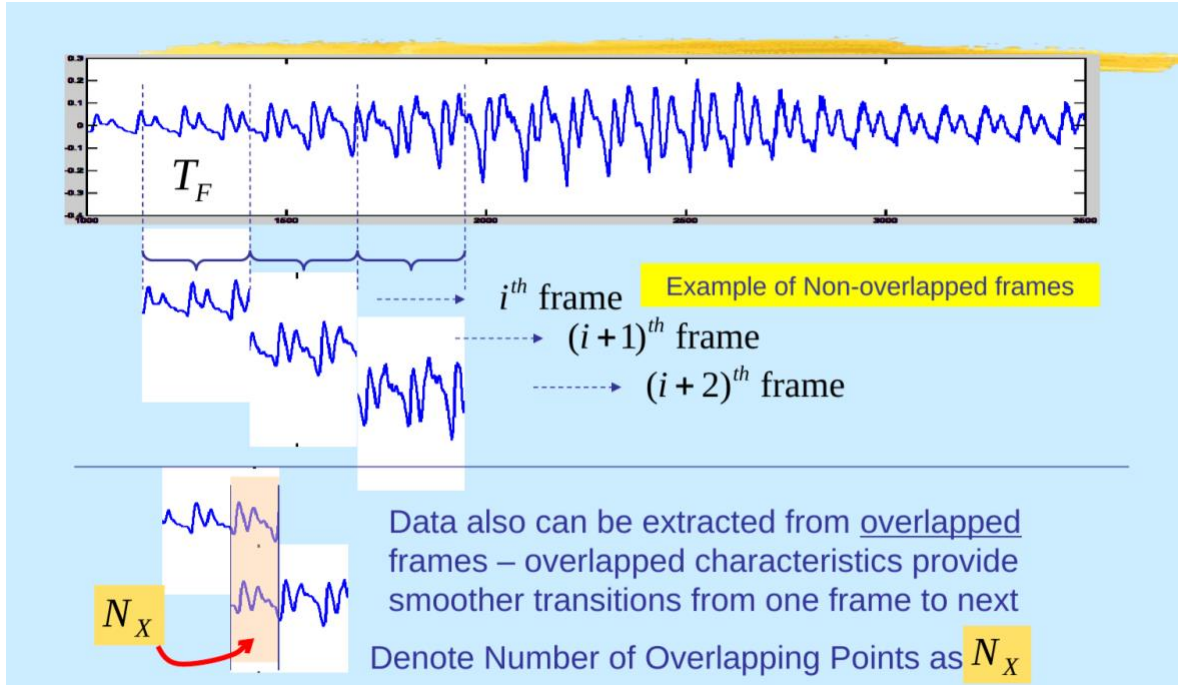


Figure 9.5. A spectrogram is built from successive short frames of the signal.

Each frame is a short windowed segment of the signal. The FFT is computed for each frame, and overlapping frames are often used to produce smoother time evolution in the spectrogram.

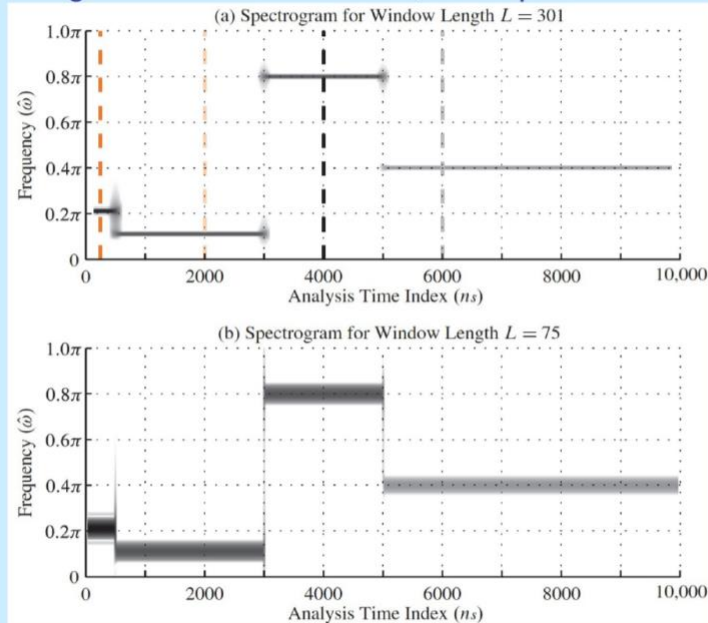
9.9.2 Spectrogram resolution

For a Hamming window of length L , a useful approximation to frequency resolution in hertz is

$$\Delta f \approx \frac{4f_s}{L}.$$

This is a mainlobe-width criterion, not merely the DFT-bin spacing f_s/N . A larger N makes the frequency grid denser; only a longer nonzero window improves the ability to separate nearby components.

■ Window Lengths of $L=301$ and $L=75$; overlap 90%



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Figure 9.6. Two spectrograms of the same signal using long and short windows.

The longer window (top) gives better frequency resolution, so the spectral lines are sharper in frequency, but time transitions are more smeared. The shorter window (bottom) gives better time resolution, but the frequency traces are broader.

9.8.3 Interpretation examples

- A periodic multi-harmonic signal produces horizontal lines at its harmonic frequencies.
- A musical scale produces a sequence of horizontal tracks that step upward in frequency; harmonics appear above each fundamental.
- Speech shows slowly varying resonances, or formants, together with harmonic structure during voiced segments.
- Filtering can be visualized by comparing input and output spectrograms.
- Short windows reveal onsets and rapid temporal changes; long windows sharpen frequency tracks but smear their start and end times.

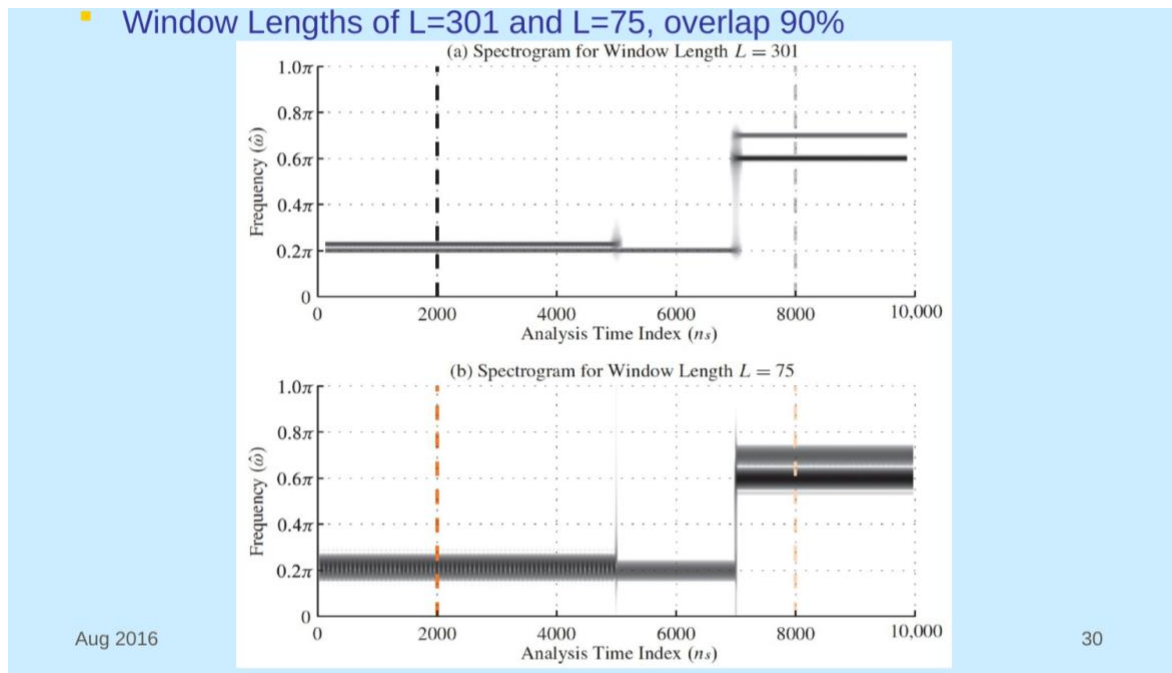


Figure 9.7. Resolution test in the spectrogram.

This example shows the classic trade-off between time and frequency resolution. The longer window separates two nearby frequency components more clearly, while the shorter window shows time changes more sharply but merges close spectral components.

9.9 Why the FFT is fast

Let

$$W_N = e^{-j2\pi/N}.$$

Then the N -point DFT can be written as

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}.$$

Splitting the samples into even and odd indices gives

$$X[k] = E[k] + W_N^k O[k],$$

and

$$X[k + N/2] = E[k] - W_N^k O[k], \quad 0 \leq k < N/2.$$

The same decomposition is applied recursively until only 2-point transforms remain. The butterfly network reuses intermediate calculations and reduces complexity from $O(N^2)$ to $O(N \log_2 N)$.

9.10 Practical workflow for reliable spectrum analysis

1. **Check sampling.** Choose f_s high enough to avoid aliasing; for real signals interpret the independent range up to $f_s/2$.
2. **Choose the segment.** Use a duration long enough for the desired frequency resolution but short enough that the signal is approximately stationary.
3. **Choose the window.** Rectangular gives a narrower mainlobe but higher sidelobes; Hamming or Hann reduces sidelobes but broadens peaks.

4. **Choose N .** Use $N \geq L$. Larger N gives denser frequency sampling but does not replace a longer observation interval.
5. **Scale the spectrum.** State whether magnitude, power, power spectral density, linear amplitude, or decibels are shown.
6. **Map bins correctly.** Use $f_k = kf_s/N$ and handle the negative-frequency half appropriately.
7. **Validate.** Test on known sinusoids and inspect the effects of bin alignment, window choice, and record length.

9.11 Common interpretation errors

Mistake	Correction
Treating FFT bins as the “true” spectrum	The FFT gives samples of the DTFT of the finite, windowed sequence.
Assuming zero-padding improves resolution	It interpolates the spectral curve; it does not reduce mainlobe width.
Ignoring window gain	Windowing changes measured amplitudes; amplitude estimates may require coherent-gain correction.
Calling every spread peak noise	A spread peak may be deterministic leakage from a sinusoid that is not bin-centred.
Using a very long window for a changing signal	Frequency detail improves, but temporal variation is averaged.
Displaying only $0 \dots f_s$ without explaining symmetry	For real signals, the independent range is normally $0 \dots f_s/2$; the remainder represents negative-frequency aliases.

9.12 Compact chapter summary

- The CTFT describes the ideal continuous-frequency spectrum of a continuous-time signal.
- Sampling creates a discrete-time signal whose DTFT is periodic and consists of shifted spectral copies.
- Finite observation is implemented by windowing, which spreads spectral components.
- The DFT samples the DTFT at N equally spaced frequencies and has an exact inverse for N samples.
- The FFT computes the DFT efficiently; it is not a different transform.
- Window duration sets the time-frequency resolution trade-off.
- Window shape controls the compromise between mainlobe width and sidelobe level.
- Zero-padding increases display density but not true resolving power.
- Repeated windowed DFTs form a spectrogram, allowing frequency content to be followed over time.

9.13 Notation

Symbol	Meaning
$x_c(t)$	Continuous-time signal
$x[n]$	Sampled discrete-time signal
T_s, f_s	Sampling period and rate, with $f_s = 1/T_s$
ω, f	Angular frequency in rad/s and cyclic frequency in Hz
$\hat{\omega}$	Normalized discrete-time angular frequency in rad/sample

Symbol	Meaning
L	Number of nonzero or windowed samples
N	DFT/FFT length
$w[n]$	Analysis window
$X[k]$	k th DFT coefficient
R	Spectrogram hop size in samples